Closing the order law's $\tau = 0$ gap: an SYT with all-free descents exists, via a descent-set Kostka number

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Abstract

The order law states $\operatorname{ord}_{x=q^2} Z_\lambda = \min_{T \in \mathrm{SYT}(\lambda)} s(T) = \tau(\tau+1)/2$ for a parity-twisted descent statistic s. Its lower bound and the $\tau > 0$ achievability were proved earlier; the lone remaining gap was $\tau = 0$ achievability — the existence of a standard Young tableau with no "paid" descent. We close it: such a tableau exists for every $\tau = 0$ shape, and in fact the number of them is a Kostka number $K_{\lambda,\mu}$, positive exactly when $\tau = 0$. The proof is the classical descent-set/Kostka identity plus a two-line dominance computation; we also correct an earlier mis-attribution of this gap to Swanson's modular major-index existence theorem.

1 Setup

Fix $n \geq 1$ and a partition $\lambda \vdash n$ with $\ell = \lambda'_1$ rows (the length of the first column). For a standard Young tableau $T \in \mathrm{SYT}(\lambda)$, the *descent set* is

$$\mathrm{Des}(T) = \{\, i \in [1, n-1] : i+1 \text{ lies in a strictly lower row than } i \text{ in } T \,\}.$$

Define the *parity-twisted weight*

$$w_i = \begin{cases} 2i-1, & n-i \text{ odd}, \\ 0, & n-i \text{ even}, \end{cases} \qquad s(T) = \sum_{i \in \mathrm{Des}(T)} w_i.$$

Call a position i *free* if $n-i$ is even (equivalently $i \equiv n \pmod 2$, so $w_i = 0$) and *paid* otherwise. Let

$$S = \{\, i \in [1, n-1] : i \equiv n \pmod 2 \,\}$$

be the set of free positions. Set

$$\tau = \max\bigl(0,\ 2\ell - n - 1\bigr).$$

Since the paid positions are exactly those carrying a nonzero weight, we have the basic equivalence

$$s(T) = 0 \iff \mathrm{Des}(T) \subseteq S \qquad \text{(all descents free).} \tag{1}$$

For background (established in the companion note `2026-05-31-orderlaw-descent-minimum`): the lower bound $s(T) \geq \tau(\tau+1)/2$ holds for all $T \in \mathrm{SYT}(\lambda)$, and for $\tau > 0$ there is an explicit minimizer attaining equality. The remaining gap was the case $\tau = 0$: does there exist T with $s(T) = 0$? By (1) this is exactly the question of whether some $T \in \mathrm{SYT}(\lambda)$ has all of its descents at free positions. We answer it affirmatively and, in fact, count such tableaux.

2 The descent-set Kostka identity

Lemma 1 (classical)**.** *For any subset $S = \{s_1 < \cdots < s_k\} \subseteq [1, n-1]$, let*

$$\alpha(S) = (s_1,\ s_2 - s_1,\ \ldots,\ s_k - s_{k-1},\ n - s_k)$$

be the composition of n whose set of partial sums (excluding the full sum n) is exactly S. Then

$$\#\{\, T \in \mathrm{SYT}(\lambda) : \mathrm{Des}(T) \subseteq S \,\} = K_{\lambda,\alpha(S)},$$

the number of semistandard Young tableaux of shape λ and content $\alpha(S)$, equivalently $\langle s_\lambda, h_{\alpha(S)}\rangle$. Since $K_{\lambda,\alpha}$ depends only on the rearrangement of α into a partition, write $\mu = \mathrm{sort}(\alpha(S))$; then the count is $K_{\lambda,\mu}$.

Proof. This is the standard descent-set/Kostka identity; see Stanley, *Enumerative Combinatorics* vol. 2, Cor. 7.19.5 and Thm. 7.19.7, and Gessel. Concretely it is the pairing of the complete-homogeneous expansion $h_\alpha = \sum_\lambda K_{\lambda\alpha}\, s_\lambda$ against the fundamental quasisymmetric expansion $s_\lambda = \sum_{T \in \mathrm{SYT}(\lambda)} F_{\mathrm{Des}(T)}$, together with the fact that the coefficient extracting $\mathrm{Des}(T) \subseteq S$ from $F_{\mathrm{Des}(T)}$ recovers precisely the tableaux whose descent set is contained in S. $\square$

3 The composition for free positions

Lemma 2. *For $S = \{\, i \in [1, n-1] : i \equiv n \pmod 2 \,\}$:*

- *if n is even: $S = \{2, 4, \ldots, n-2\}$, so $\alpha(S) = (2, 2, \ldots, 2)$ with $n/2$ twos, and $\mu = (2^{n/2})$;*
- *if n is odd: $S = \{1, 3, \ldots, n-2\}$, so $\alpha(S) = (1, 2, 2, \ldots, 2)$ (a single 1 followed by $(n-1)/2$ twos), and $\mu = (2^{(n-1)/2}, 1)$.*

In both cases the conjugate partition is

$$\mu' = \big(\lceil n/2 \rceil,\ \lfloor n/2 \rfloor\big),$$

a partition with at most two parts; in particular $\mu'_1 = \lceil n/2 \rceil$.

Proof. We read off $\alpha(S)$ from the partial-sum description of Lemma 1.

n even. The free positions are the even integers in $[1, n-1]$, namely $S = \{2, 4, \ldots, n-2\}$. The consecutive gaps are all 2, and the final gap is $n - (n-2) = 2$, so $\alpha(S) = (2, 2, \ldots, 2)$ with $n/2$ parts. Its partial sums are $2, 4, \ldots, n$, and intersecting with $[1, n-1]$ recovers S, as required. Sorting gives $\mu = (2^{n/2})$.

n odd. The free positions are the odd integers in $[1, n-1]$, namely $S = \{1, 3, \ldots, n-2\}$. The first gap is 1, the remaining consecutive gaps are 2, and the final gap is $n - (n-2) = 2$, so $\alpha(S) = (1, 2, 2, \ldots, 2)$ with a leading 1 and $(n-1)/2$ twos. Its partial sums are $1, 3, 5, \ldots, n$, whose intersection with $[1, n-1]$ is S. Sorting gives $\mu = (2^{(n-1)/2}, 1)$.

For the conjugate, count column lengths. In either case μ has parts equal to 2 and possibly one part equal to 1. The first column of μ has one box per part, i.e. μ'_1 equals the number of parts: $n/2$ when n is even and $(n-1)/2 + 1 = (n+1)/2$ when n is odd; both equal $\lceil n/2 \rceil$. The second column has one box for each part of size ≥ 2, i.e. μ'_2 equals the number of twos: $n/2$ when n is even and $(n-1)/2$ when n is odd; both equal $\lfloor n/2 \rfloor$. There is no third column. Hence $\mu' = (\lceil n/2 \rceil, \lfloor n/2 \rfloor)$. $\square$

4 Dominance

We write $\trianglerighteq$ for the dominance order on partitions of n.

Lemma 3. *With μ as in Lemma 2,*

$$K_{\lambda,\mu} > 0 \iff \lambda \trianglerighteq \mu \iff \mu' \trianglerighteq \lambda' \iff \lambda'_1 \leq \lceil n/2 \rceil \iff \tau(\lambda) = 0.$$

Proof. The first equivalence is the Kostka positivity criterion: $K_{\lambda\mu} > 0$ if and only if λ dominates μ. The second is the standard fact that dominance is reversed under conjugation: $\lambda \trianglerighteq \mu \iff \mu' \trianglerighteq \lambda'$.

For the third equivalence, recall from Lemma 2 that $\mu' = (\lceil n/2 \rceil, \lfloor n/2 \rfloor)$ has only two (possibly nonzero) parts. The dominance condition $\mu' \trianglerighteq \lambda'$ requires

$$\sum_{i \le j} \mu'_i \;\ge\; \sum_{i \le j} \lambda'_i \qquad \text{for all } j \ge 1.$$

For $j = 1$ this reads $\lceil n/2 \rceil \ge \lambda'_1$. For $j \ge 2$ the left-hand side is already the full sum $\mu'_1 + \mu'_2 = \lceil n/2 \rceil + \lfloor n/2 \rfloor = n$, while the right-hand side $\sum_{i \le j} \lambda'_i \le n$; the inequality is therefore automatic. Hence the single binding constraint is $\lambda'_1 \le \lceil n/2 \rceil$, i.e. $\ell \le \lceil n/2 \rceil$.

Finally, $\lambda'_1 \le \lceil n/2 \rceil \iff 2\lambda'_1 \le 2\lceil n/2 \rceil$. Since $2\lceil n/2 \rceil \le n+1$ for both parities (equality when n is odd, and $2\lceil n/2 \rceil = n \le n+1$ when n is even), and since λ'_1 is an integer, $\lambda'_1 \le \lceil n/2 \rceil$ is equivalent to $2\lambda'_1 \le n+1$, i.e. to $2\lambda'_1 - n - 1 \le 0$, i.e. to $\tau = \max(0, 2\lambda'_1 - n - 1) = 0$. □

5 Main theorem

Theorem 1. *Let $\lambda \vdash n$ with $\tau(\lambda) = 0$, and let μ be as in Lemma 2. Then*

$$\#\{\, T \in \mathrm{SYT}(\lambda) : s(T) = 0 \,\} = K_{\lambda,\mu} \;\ge\; 1.$$

In particular there exists $T \in \mathrm{SYT}(\lambda)$ achieving $s(T) = 0 = \tau(\tau+1)/2$. Combined with the lower bound $s(T) \ge \tau(\tau+1)/2$ and the explicit $\tau > 0$ construction, the order law

$$\min_{T \in \mathrm{SYT}(\lambda)} s(T) = \tfrac{\tau(\tau+1)}{2}$$

now holds, with full rigor, uniformly over all partitions λ.

Proof. By (1), $s(T) = 0$ if and only if $\mathrm{Des}(T) \subseteq S$. Applying Lemma 1 with this S counts such tableaux as $K_{\lambda,\alpha(S)} = K_{\lambda,\mu}$, where Lemma 2 identifies μ. Since $\tau(\lambda) = 0$, Lemma 3 gives $K_{\lambda,\mu} > 0$, hence the count is at least 1 and a minimizer with $s(T) = 0$ exists. The remaining clause assembles the three legs: the lower bound and $\tau > 0$ achievability are established in the companion note, and the present theorem supplies $\tau = 0$ achievability, so the minimum equals $\tau(\tau+1)/2$ for every λ. □

6 Remarks

Remark 1 (Exact count of minimizers). For $\tau = 0$ the number of order-law minimizers is precisely the Kostka number $K_{\lambda,\mu}$ — a refinement of mere existence. For instance, with $n = 7$: the shape $\lambda = (4, 2, 1)$ gives $K_{\lambda,\mu} = 6$, and $\lambda = (5, 2)$ gives $K_{\lambda,\mu} = 5$.

Remark 2 (Conjugate form). Under transposition one has $s(T') = \binom{n}{2} - s(T)$. Consequently the count above can be read on the conjugate shape as

$$\#\{\, T' \in \mathrm{SYT}(\lambda') : \text{every paid position is a descent} \,\} = K_{\lambda,\mu}.$$

Remark 3 (Correction of attribution). This gap was previously conjectured to coincide with Swanson's existence theorem for tableaux with a prescribed modular major index (arXiv:1701.04963). It does not. Swanson controls

$$\mathrm{maj}(T) = \sum_{i \in \mathrm{Des}(T)} i \pmod{n},$$

a single linear functional of the descent set, reduced modulo n. The condition $s(T) = 0$ instead constrains *which positions, by parity*, may be descents. The latter is a finer, positional condition on the descent set that Swanson's mod-n machinery cannot detect: a tableau can realize any target value of maj mod n while still placing a descent at a wrong-parity (paid) position. The correct and far more elementary tool is the descent-set Kostka positivity of Lemma 1.

7 The exact minimizer count and uniform achievability

The $\tau = 0$ argument extends verbatim to all τ and pins down the exact number of order-law minimizers, replacing the explicit $\tau > 0$ construction of the companion note with the same one-line dominance fact.

Write $\ell = \lambda'_1$ (the number of rows) and $K = \lceil \tau/2 \rceil$. Recall that the paid positions are those i with $n - i$ odd, the free positions those with $n - i$ even (equivalently $i \equiv n \pmod 2$); let F denote the set of free positions in $[1, n-1]$. Let $p_1 < p_2 < \cdots$ enumerate the paid positions; the weights $w_{p_1} < w_{p_2} < \cdots$ increase. Define

$$S^* = F \cup \{p_1, \ldots, p_K\} \qquad \text{(free positions together with the } K \text{ cheapest paid positions).}$$

Proposition 1 (Minimizers are exactly the $\mathrm{Des} \subseteq S^*$ tableaux)**.** *For* $T \in \mathrm{SYT}(\lambda)$, $s(T) = \tau(\tau+1)/2 \iff \mathrm{Des}(T) \subseteq S^*$.

Proof. ($\Leftarrow$) If $\mathrm{Des}(T) \subseteq S^*$ then every paid descent lies in $\{p_1, \ldots, p_K\}$, so T has at most K paid descents; the lower-bound lemma (Lemma B of the companion note) forces at least K. Hence T has exactly K paid descents, necessarily at $p_1, \ldots, p_K$, and $s(T) = \sum_{t=1}^{K} w_{p_t} = \tau(\tau+1)/2$.

($\Rightarrow$) Let $P = \{i_1 < i_2 < \cdots\}$ be the paid descents of T. The lower-bound chain

$$s(T) = \sum_{t \le |P|} w_{i_t} \;\ge\; \sum_{t \le |P|} w_{p_t} \;\ge\; \sum_{t \le K} w_{p_t} = \tfrac{\tau(\tau+1)}{2}$$

consists entirely of equalities when $s(T)$ is minimal. Equality in the second step forces $|P| = K$; equality in the first forces $i_t = p_t$ for $t \le K$. So the paid descents are exactly $\{p_1, \ldots, p_K\} \subseteq S^*$, and the free descents lie in $F \subseteq S^*$; thus $\mathrm{Des}(T) \subseteq S^*$. $\square$

Theorem 2 (Exact minimizer count; uniform achievability)**.** *Let* $M(\lambda) = \#\{T \in \mathrm{SYT}(\lambda) : s(T) = \tau(\tau+1)/2\}$. *Then* $M(\lambda) = K_{\lambda,\nu}$, *where for* $\tau > 0$

$$\nu = (2^{\,n-\ell},\, 1^{\,\tau+1}), \qquad \nu' = (\ell,\, n-\ell),$$

and for $\tau = 0$, $\nu = \mu$ *of the previous sections. In every case* $\lambda \trianglerighteq \nu$, *so* $M(\lambda) \ge 1$*: the order-law minimum is achieved for all* λ, *with no explicit construction.*

Proof. By the Proposition and Lemma 1 (the descent-set Kostka identity), $M(\lambda) = \#\{T : \mathrm{Des}(T) \subseteq S^*\} = K_{\lambda,\alpha(S^*)}$. We compute $\alpha(S^*)$ for $\tau > 0$. Since $\tau + 1 = 2\ell - n \equiv n \pmod 2$, the position $\tau + 1$ is free. One checks (in both parities of n) that the $K = \lceil \tau/2 \rceil$ cheapest paid positions are exactly the paid positions $\le \tau$, so

$$S^* \;=\; \{1, 2, \ldots, \tau+1\} \;\cup\; \{\text{free positions in } [\tau+2,\, n-1]\},$$

and the free positions in $[\tau+2, n-1]$ are $\{\tau+3, \tau+5, \ldots, n-2\}$. Thus the sorted S^* is $1, 2, \ldots, \tau{+}1,\ \tau{+}3, \tau{+}5, \ldots, n{-}2$, whose consecutive gaps give the composition $\alpha(S^*) = (1^{\,\tau+1}, 2^{\,n-\ell})$: the initial run $1, \ldots, \tau{+}1$ contributes $\tau{+}1$ parts equal to 1, and the subsequent step-2 jumps together with the final part $n - (n-2) = 2$ contribute $\frac{n-1-\tau}{2} = n - \ell$ parts equal to 2. Rearranged, this is $\nu = (2^{\,n-\ell}, 1^{\,\tau+1})$, of size $(\tau{+}1) {+} 2(n{-}\ell) = n$ (using $\tau{+}1 = 2\ell{-}n$). Its conjugate is $\nu' = (\ell,\, n-\ell)$, a two-part partition. Hence $\lambda \trianglerighteq \nu \iff \nu' \trianglerighteq \lambda'$, whose only binding inequality is $\nu'_1 = \ell \ge \lambda'_1 = \ell$ — an equality, hence always satisfied (the $j \geq 2$ inequalities read $n \geq \sum_{i \leq j} \lambda'_i$). Therefore $K_{\lambda,\nu} \geq 1$. For $\tau = 0$ the count and the dominance reduce to the previous sections ($\nu'_1 = \lceil n/2 \rceil \geq \ell \iff \tau = 0$). In all cases $M(\lambda) = K_{\lambda,\nu} \geq 1$, giving a construction-free, uniform proof of achievability that subsumes the explicit $\tau > 0$ construction of the companion note. $\square$

Remark 4. The content ν is always of two-column shape (parts ≤ 2): $\nu = (2^{n-\ell}, 1^{\tau+1})$ for $\tau > 0$ and $(2^{\lfloor n/2 \rfloor}, 1^{n-2\lfloor n/2 \rfloor})$ for $\tau = 0$. The number of order-law minimizers is therefore a Kostka number counting SSYT of two-column content — e.g. $n = 7$: $\lambda = (4, 2, 1)$ gives $M = 6$, $\lambda = (5, 2)$ gives $M = 5$ (both $\tau = 0$); for $\tau > 0$ shapes M is typically 1. There is *no* conjugation symmetry $M(\lambda) = M(\lambda')$, since $\tau(\lambda) \neq \tau(\lambda')$ in general.

8 Computational verification

The minimizer-count identity $M(\lambda) = K_{\lambda,\nu}$ with $\nu = (2^{n-\ell}, 1^{\tau+1})$ was verified with 0 mismatches over all 194 partitions $n \leq 11$ via independent SSYT enumeration (`verify_kostka_bridge.py` / `min_mult_task2.py`); the equivalence "minimizers $\iff$ $\mathrm{Des} \subseteq S^*$" was checked with 0/1979 exceptions. All three legs were verified by an independent script `verify_kostka_bridge.py`, which enumerates semistandard tableaux for the Kostka side using no shared code with the SYT/$s(T)$ side:

(i) $\#\{T : s(T) = 0\} = K_{\lambda,\mu}$: **0 mismatches** over all 194 partitions with $n \leq 11$.

(ii) $\lambda \trianglerighteq \mu \iff \tau = 0$: **0 failures** over all 507 partitions with $n \leq 14$.

(iii) the single binding constraint $\ell \leq \lceil n/2 \rceil$: **0 failures** for $n \leq 12$.

Earlier, all $\tau = 0$ shapes with $n \leq 12$ were brute-confirmed to admit an explicit $s(T) = 0$ witness, in agreement with the existence statement of Theorem 1.